



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

DATA STRUCTURE & ALGORITHMS (SECJ2013)

SEMESTER 1 2024/2025

**Mini Project Documentation:
Convocation Ceremony System**

By SECTION 02

**LECTURER:
Dr Zuraini binti Ali Shah**

Name	Matric ID
LEE YIN SHEN	A23CS0236
AHMAD ZIYAAD BIN MOHD ABBAS	A23CS0206
ELIJAH SHE YU SHENG	A23CS0073
NURUL ASYIKIN BINTI KHAIRUL ANUAR	A23CS0162

Table of Content

PART 1: INTRODUCTION

1.1	Synopsis Project	3
1.2	Objective of the Project	4

PART 2: SYSTEM ANALYSIS AND DESIGN (USE CASE AND CLASS DIAGRAM)

2.1	System Requirement -Use Case Diagram	5
2.2	System Design - UML Class Diagram	7

PART 3: SYSTEM PROTOTYPE

		8
--	--	---

PART 4: Task Distribution and Log Sheet

		15
--	--	----

Conclusion

		17
--	--	----

Reference

		18
--	--	----

PART 1: INTRODUCTION

1.1 Synopsis Project

The UTM Convocation Ceremony System is a comprehensive digital approach for streamlining and managing the graduation ceremony events at Universiti Teknologi Malaysia (UTM). This system attempts to improve efficiency and organize the convocation process by managing graduate registration, ceremony flow and real-time queue monitoring. With an increasing number of graduates each year, a systematic and efficient management system is critical to ensuring that the ceremonies run well. The system keeps complete records, including the personal information, academic achievements and ceremonial requirements, all accessible via a user-friendly interface that ensures data accuracy and accessibility.

The system uses basic data structures, primarily queues to control the sequential flow of the graduates during the ceremony. It offers the staff capabilities of registering graduates, maintaining their academic records and overseeing the orderly delivery of certificates during the convocation event. The system maintains a smooth flow of graduates using its real-time queue management feature, which automatically tracks their progress and provides quick updates to ceremony coordinators. This systematic approach maximizes administrative effectiveness while improving each participant's ceremonial experience.

1.2 Objective of the Project

The primary objective of the Convocation Ceremony System are:

1. To properly construct the data structures and algorithm in order to create an effective digital system that simplifies the management of UTM's convocation ceremonies.
2. To develop an intuitive user interface that enables staff members to efficiently register and manage graduation data such as academic accomplishments, personal information and status as a ceremony participant.
3. Implement a queue-based system to guarantee the orderly movement of graduates throughout the ceremony, as well as right sequence and timing for certificate distribution.
4. Provide real-time monitoring options for staff to track the ceremony progress including the ability to see both the first and last graduates in the line at any given time.
5. To improve the overall efficiency of the convocation event by eliminating manual administrative tasks and decreasing potential organizational errors through systematic digital management.

PART 2: SYSTEM ANALYSIS AND DESIGN (USE CASE AND CLASS DIAGRAM)

2.1. System Requirements

Use case diagram – Describe the users of the system and detailed descriptions of each use case.

Example:

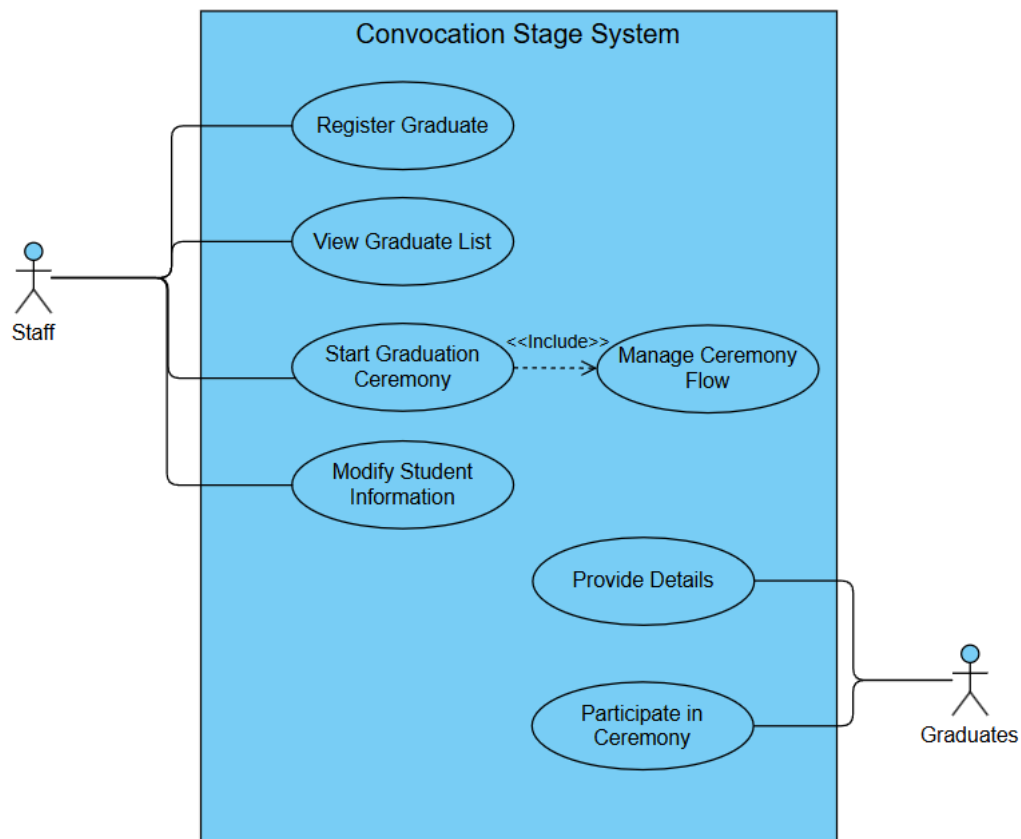


Figure 1: Use Case Diagram for Convocation Ceremony System.

Use Case Description for UTM Convocation Ceremony System

In the UTM Convocation Ceremony System, the primary system users are staff and graduates.

Actor	Task
Staff Member	Administrative personnel responsible for managing all aspects of the convocation ceremony, including registering graduates, overseeing the ceremony proceedings, and maintaining records.
Graduates	Students participating in the graduation ceremony. They provide their details (name, course, GPA). They also participate in the ceremony.

Table 1: Use Case Description for Convocation Ceremony System

Use Case Description for UTM Convocation Ceremony System

Use case	Purpose
Register Graduate	Allows staff members to add new graduates to the system by entering their personal and academic details, ensuring accurate and up-to-date records.
View Graduate List	Enables staff to view a comprehensive list of all registered graduates, including their names, courses, GPAs, and statuses (exp: Dean's List).
Start Convocation Ceremony	Initiates the convocation ceremony process, managing the sequence in which graduates receive their certificates and ensuring a smooth flow of the event.
Manage Ceremony Flow	Oversees the progression of the convocation ceremony by handling the arrival of graduates and the distribution of certificates. Moreover, staff is able to see the first and also the last graduates in the queue, maintaining real-time status updates.

Table 2: Detail Description for Each Use Case

2.2. System Design

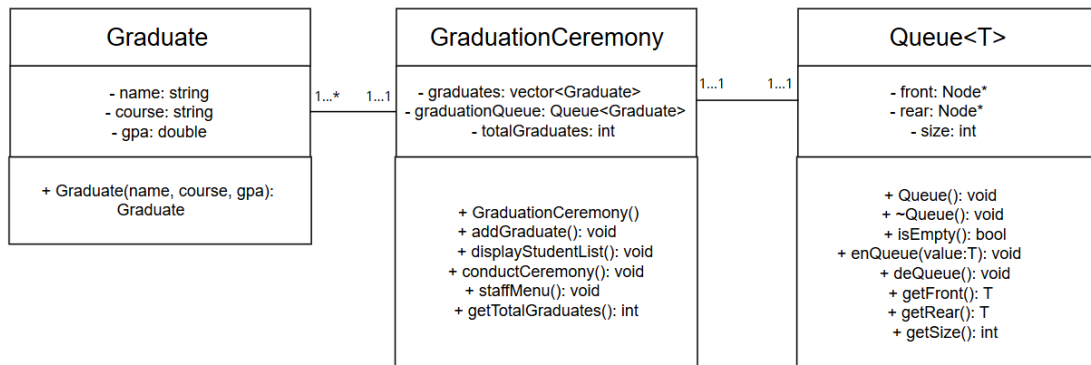


Figure 2: UML Class Diagram for Convocation Ceremony System.

Classes and Relationships with Multiplicity

1. GraduationCeremony → Graduate

Relationship: uses

Multiplicity: 1...1 → 1..*

Explanation: The GraduationCeremony class uses many (1..*) Graduate objects stored in a vector. At any point, a GraduationCeremony may have one or more graduates registered.

2. GraduationCeremony → Queue<Graduate>

Relationship: has-a

Multiplicity: 1...1 → 1...1

Explanation: The GraduationCeremony class has exactly one (1) Queue<Graduate> object, which is used to manage the ceremony. The queue always exists as part of the ceremony.

PART 3: SYSTEM PROTOTYPE

1. Graduation Ceremony System Menu

```
=== Graduation Ceremony System ===
1. View Student List
2. Add Student
3. Start Convocation Ceremony
4. Exit
Enter choice: |
```

There are four options in the main menu, 1 is for viewing the student list, where a user can view both registered graduates in the system. 2 is for adding or registering graduates into the system. 3 is for starting the convocation ceremony simulation. While 4 is to exit the system.

2. View Student List

```
=== Graduation Ceremony System ===
1. View Student List
2. Add Student
3. Start Convocation Ceremony
4. Exit
Enter choice: 1

Registered Students:
-----
No.  Name                Course                GPA    Status
-----
1    Lee Yin Shen          Software Engineering   3.80   Dean's List
2    Haikal Bin Abdullah    Software Engineering   3.22   Dean's List
3    Syafiq                 Software Engineering   3.99   Dean's List
-----
```

For option 1 - view student list, user can view the list of registered graduates in the system. The name, course, GPA and dean's list status of a graduate will be displayed.

3. Add Student

```
=== Graduation Ceremony System ===
1. View Student List
2. Add Student
3. Start Convocation Ceremony
4. Exit
Enter choice: 2

Enter graduate details
-----
Name: Lee Yin Shen

Select Course:
1 - Data Engineering
2 - Software Engineering
3 - Computer Networks & Security
4 - Graphics and Multimedia Software
Enter choice (1-4): 1

GPA (0.00-4.00): 3.88

Student successfully registered!

Do you want to add another student? (Y/N): |
```

For option 2 - Add Student, users can enter the graduate details, including name, course and GPA. After successfully registering a graduate, the system will prompt if a user wants to add a new graduate. If enter “Y”, the user can register a new graduate, while entering “N” will turn back to the main menu.

4. Start Convocation Ceremony

```
=== Graduation Ceremony System ===
1. View Student List
2. Add Student
3. Start Convocation Ceremony
4. Exit
Enter choice: 3

===Faculty of Computing Graduation Ceremony===
Total number of graduates: 5

Waiting Area:
-----
Empty
-----

'N' - call the next graduate to arrive
'F' - view first graduate
'R' - view last graduate

Enter choice: |
```

For option 3 - Start Convocation Ceremony, the system will show a simple simulation of the convocation ceremony. It will initially show an empty waiting area. There are three options for managing the ceremony, “N” is calling the next graduate to arrive, “F” is to view the first student in the queue, while “R” is to view the last student in the queue.

```
Enter choice: n

Waiting Area:
-----
Lee Yin Shen (1)
-----

'N' - call the next graduate to arrive
'F' - view first graduate
'R' - view last graduate

Enter choice: n

Waiting Area:
-----
Lee Yin Shen (1)
Haikal (2)
-----

'N' - call the next graduate to arrive
'F' - view first graduate
'R' - view last graduate

Enter choice: n

Waiting Area:
-----
Lee Yin Shen (1)
Haikal (2)
Shilla (3)
-----
```

Users can enter “N” to call the graduate to arrive at the waiting area one by one until all graduates arrive.

```
'N' - call the next graduate to arrive
'F' - view first graduate
'R' - view last graduate

Enter choice: F

First graduate in waiting area: Lee Yin Shen (1)

Waiting Area:
-----
Lee Yin Shen (1)
Haikal (2)
Shilla (3)
Aiman (4)
-----
```

```
Enter choice: R

Last graduate in waiting area: Aiman (4)

Waiting Area:
-----
Lee Yin Shen (1)
Haikal (2)
Shilla (3)
Aiman (4)
-----
```

During the arrival of graduates, users can enter “F” to view the first student in the current queue, “R” to view the last student in the current queue.

```

All graduates have arrived!

=== Starting Certificate Ceremony ===

Current Status:
-----
Waiting Area                               Finished Area:
-----
Lee Yin Shen (1)
Haikal (2)
Shilla (3)
Aiman (4)
Xu Xin (5)
-----

'N' - call the next graduate to receive cert
'F' - view first graduate
'R' - view last graduate
'Q' - quit

Enter choice: |

```

When all graduates arrive, the certificate ceremony will start. There are four options now, “N” is to call the next graduate to receive a certificate, “F” is to view the first graduate in the current queue, “R” is to view the last graduate in the current queue, while “Q” is to end the ceremony.

```

Enter choice: n

* Lee Yin Shen (1) has received their certificate *

Current Status:
-----
Waiting Area                               Finished Area:
-----
Haikal (2)                                Lee Yin Shen (1)
Shilla (3)
Aiman (4)
Xu Xin (5)
-----

```

If you enter “N”, the first graduate who arrives at the waiting area will receive the certificate and move to the finished area first. By repeatedly entering “N”, the rest of the graduates will receive and move to the finished area by sequence until both graduates receive their certificate.

```

Enter choice: f
First graduate in waiting area: Shilla (3)
Current Status:
-----
Waiting Area                               Finished Area:
-----
Shilla (3)                                Lee Yin Shen (1)
Aiman (4)                                Haikal (2)
Xu Xin (5)
-----

```

```

Enter choice: r
Last graduate in waiting area: Xu Xin (5)
Current Status:
-----
Waiting Area                               Finished Area:
-----
Shilla (3)                                Lee Yin Shen (1)
Aiman (4)                                Haikal (2)
Xu Xin (5)
-----

```

During this process, users can also enter “F” to view the first student in the current queue, “R” to view the last student in the current queue.

```

Ceremony finished! Congratulations to all graduates!

```

After all graduates receive their certificate, the convocation ceremony will be finished. Users enter option 4 to exit the system.

PART 4: Task Distribution and Log Sheet

Task Distribution

Name	Matric ID	Responsibilities
Lee Yin Shen	A23CS0236	Develop prototype system menu and implement graduate registration functionality. Code queue system for graduate sequences and integrate real-time tracking.
Ahmad Ziyaad	A23CS0206	Document task distribution table and create a log sheet. Write a conclusion summarizing project objectives, outcomes, and references.
Elijah She Yu Sheng	A23CS0073	Create use cases and class diagrams, and write descriptions for system interactions.
Nurul Asyikin	A23CS0162	Write project introduction, objectives, and manage system synopsis and requirements descriptions.

Log Sheet

Date	Task	Assigned Member(s)	Details	Status
10/12/2025	Initial Project Planning	All Members	Discuss project scope, finalize task allocation, and determine submission guidelines.	Completed
10/12/2025	Write Project Introduction & Objectives	Nurul Asyikin	Prepare the introductory section and clarify the system's goals.	Completed
11/12/2025	Prototype System Menu Design	Lee Yin Shen	Implement core functionalities for the convocation ceremony system, including queue simulation.	Completed
11/12/2025	Create Use Case and Class Diagrams	Elijah She Yu Sheng	Finalize system design diagrams and add detailed descriptions.	Completed
12/12/2025	Task Distribution and Log Sheet Documentation	Ahmad Ziyaad	Complete the log sheet and prepare the conclusion and references.	Completed
12/12/2025	Final Review and Report Compilation	All Members	Conduct a final review of all components and compile the documentation for submission.	Completed

Conclusion

The queue data structure is the backbone of the Convocation Ceremony System, ensuring an orderly and efficient flow of graduates throughout the event. In this system, a FIFO (First-In-First-Out) mechanism is employed, mirroring the natural sequence of a graduation ceremony[1]. Graduates are added to the queue as they register and proceed through the ceremony in the same order, maintaining fairness and precision.

This structure is particularly appropriate for managing events that require sequential processes, such as certificate distribution and tracking ceremony progress[2]. By leveraging a queue, the system enables real-time tracking of the first and last graduates in line, ensuring staff can efficiently monitor and manage the event.

The queue's simplicity, combined with its ability to handle dynamic changes, makes it a perfect choice for this project. It ensures scalability and reliability, demonstrating the power of fundamental data structures in solving practical, event-management challenges[3].

References

1. GeeksforGeeks. (n.d.). *Applications of Queue Data Structure*. Retrieved from <https://www.geeksforgeeks.org/applications-of-queue-data-structure>
2. Qminder. (n.d.). *Make Your Event Successful With a Queue Management System*. Retrieved from <https://www.qminder.com/blog/queue-management/event-queue-management-system>
3. Shiksha. (n.d.). *Queue Data Structure: Types, Implementation, Applications*. Retrieved from <https://www.shiksha.com/online-courses/articles/queue-data-structure-types-implementation-applications>